MACHINE LEARNING

Classification Metrics

Confusion Matrix · Precision · Recall · F1 Score

A structured guide to evaluating ML models

01 Confusion Matrix

Definition: A table used to evaluate classification models by comparing **actual vs predicted values** — giving a full picture of where the model succeeds and fails.

	Predicted YES	Predicted NO
Actual YES	**TP** True Positive	**FN** False Negative
Actual NO	**FP** False Positive	**TN** True Negative

Why use it?

- Full performance picture
- Understand model errors
- Better than accuracy alone

02 Precision & Recall

🎯 Precision

TP / (TP + FP)

Of all the items predicted as positive — how many were actually positive?

Predicted 10 positives, 8 correct → Precision = 0.80

Key insight: Minimises false alarms

🔍 Recall

TP / (TP + FN)

Of all actual positives — how many did the model successfully detect?

10 actual positives, found 8 → Recall = 0.80

Key insight: Maximises true captures

03 F1 Score

The Harmonic Mean of Precision & Recall

F1 penalises extreme differences between Precision and Recall — you need both to be high.

```
F1 = 2 × (Precision × Recall) / (Precision + Recall)
```

ranges from 0 (worst) to 1 (best)

P=0.90 R=0.10

F1 = 0.18

High precision, low recall → misses most positives

P=0.80 R=0.80

F1 = 0.80

Both balanced → strong model

P=0.10 R=0.90

F1 = 0.18

High recall, low precision → too many false alarms

04 Macro & Weighted Average

Macro Average

1. Calculate metric for each class
2. Take simple (unweighted) average
3. All classes treated equally

Best when: classes are balanced in size

Treats a rare class the same as a common one

Weighted Average

1. Calculate metric for each class
2. Weight by number of samples per class
3. Common classes influence more

Best when: dataset is imbalanced

Reflects real-world performance more accurately

Key Takeaways

Confusion Matrix

Full performance picture — see TP, TN, FP, FN at a glance

Precision

Correctness of positive predictions — reduces false alarms

Recall

Coverage of actual positives — minimises missed cases

F1 Score

Harmonic balance between Precision and Recall

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